4-1-26 (WEEK 6)

Shun/翔海 (@shun4midx)

THEOREM (DEDEKIND-KUMMER)

Fix a max ideal $p \trianglelefteq \mathcal{O}_K$, and choose $\alpha \in \mathcal{O}_L$, s.t. $L = K(\alpha)$.
Let $\Phi_{\alpha,K}(x) \in \mathcal{O}_K[x]$ be the minimal polynomial of α.
Let $\overline{\Phi}_{\alpha,K} \in \overset{\mathcal{O}_K/p}{\kappa(p)}(x)$ be the reduction of $\Phi_{\alpha,K}$ mod p, and write $\overline{\Phi}_{\alpha,K} = \prod_{i=1}^{n} \overline{\varphi}_i^{e_i}$ with $\overline{\varphi}_i$ being distinct irr poly in $\kappa(p)[x]$. $\forall i$, say $\varphi_i \in \mathcal{O}_K[x]$ is a monic lifting of $\overline{\varphi}_i$.

Suppose $p \nmid \operatorname{disc}(\Phi_{\alpha,K})\, \delta(\mathcal{O}_L/\mathcal{O}_K)^{-1}$. Then,
(i) $\mathcal{O}_{K,p}[\alpha] = \mathcal{O}_{L,p}$ (Recall: $\mathcal{O}_{K,p} = (\mathcal{O}_K)_p$)
(ii) $p\mathcal{O}_L = \prod_{i=1}^{n} B_i^{e_i}$, where $B_i = p\mathcal{O}_L + \varphi_i(\alpha)\mathcal{O}_L$ and $\underbrace{f(B_i|p)}_{=[\mathcal{O}_L/B_i : \mathcal{O}_K/p]} = \deg \varphi_i$ for $1 \le i \le n$

Proof

(i) (Recall: $\operatorname{disc}(\Phi_{n,K}) = \delta(\mathcal{O}_K[\alpha]/\mathcal{O}_K)$, $\mathcal{O}_{L,p} = S^{-1}\mathcal{O}_L$, $S = \mathcal{O}_K \setminus p$)
("$\Leftrightarrow$" $\mathcal{O}_K[\alpha] = \mathcal{O}_L$)
As $p \nmid \delta(\mathcal{O}_K[\alpha]/\mathcal{O}_K)\,\delta(\mathcal{O}_L/\mathcal{O}_K)^{-1}$, $S = \mathcal{O}_K \setminus p$, $S^{-1}\delta(\mathcal{O}_K[\alpha]/\mathcal{O}_K) = S^{-1}\delta(\mathcal{O}_L/\mathcal{O}_K)$
By lemma, $\delta(S^{-1}\mathcal{O}_K[\alpha]/S^{-1}\mathcal{O}_K) = \delta(\mathcal{O}_{L,p}/\mathcal{O}_{K,p})$, so by prop, $S^{-1}\mathcal{O}_K[\alpha] = \mathcal{O}_{L,p}$ ✓

(ii) We have $\mathcal{O}_L/p\mathcal{O}_L = \mathcal{O}_{L,p}/p\mathcal{O}_{L,p} = \mathcal{O}_{K,p}[x]/(p, \Phi_{\alpha,K}) = \kappa(p)[x]/\overline{\Phi}_{\alpha,K} = \prod_{i=1}^{n} \kappa(p)[x]/\overline{\varphi}_i^{e_i}$

Thus, the prime factors of $p\mathcal{O}_L$ are $B_i = (\varphi_i(\alpha), p)$, and we can write $p\mathcal{O}_L = \prod_{i=1}^{n} B_i^{\varepsilon_i}$, $\varepsilon_i \in \mathbb{N}$

Moreover, $\prod_{i=1}^{n} B_i^{\varepsilon_i} = 0$ in $\mathcal{O}_L/p\mathcal{O}_L = \prod_{i=1}^{n} \kappa(p)[x]/\overline{\varphi}_i^{e_i}$
$\therefore$ $\varepsilon_i \ge e_i$ and $\prod_{i=1}^{n} B_i^{e_i} \subseteq p\mathcal{O}_L$
$\therefore$ $\prod_{i=1}^{n} B_i^{e_i} = p\mathcal{O}_L$ □

CORRECTION

$P \trianglelefteq \mathcal{O}_K$: max is unramified in $\mathcal{O}_L$ if
① $P\mathcal{O}_L = \prod_{i=1}^{n} B_i$, B_i: distinct + max
② $\kappa(B_i)/\kappa(p)$ is separable (auto if number field, but not for DDK, $\kappa(B_i) = \mathcal{O}_L/B_i$)

COROLLARY

$\Phi_{\alpha,K}$ is separable, then $p\mathcal{O}_L = \prod_{i=1}^{n} B_i$

EXAMPLE

$f(t) = t^3 - t - 1$, $\mathcal{O}_K = \mathbb{Z}$, $L = \mathbb{Q}[x]/(f(t))$
We recall $\operatorname{disc}(f) = \operatorname{disc}(\mathcal{O}_K[\alpha]/\mathcal{O}_K) = -23$, $f(\alpha) = 0 \Rightarrow \mathcal{O}_K[\alpha] = \mathcal{O}_L$

For any $P \trianglelefteq \mathcal{O}_K$: max, we can apply D-K Thm
$p = 23$, $\overline{f} = (t-3)(t-10)^2$ in $\mathbb{F}_{23}[t] \Rightarrow p\mathcal{O}_L = (23, \alpha-3)(23, \alpha-10)^2$

For a prime $p \neq 23$ in $\mathbb{Z}$, $p \nmid \operatorname{disc}(f) \Rightarrow f$: sep in $\mathbb{F}_p[x]$
By Coro, p: unramified

$\therefore$ $p = 23$ is the only ramified prime in $\mathcal{O}_L$.

THEOREM

Let $p \trianglelefteq \mathcal{O}_K$: max. Then, p: ramified in $\mathcal{O}_L \Leftrightarrow p \mid \delta(\mathcal{O}_L/\mathcal{O}_K)$

Proof

We know $p\mathcal{O}_L = \prod_{i=1}^{n} B_i^{e_i}$, B_i: distinct max
$\mathcal{O}_L/p\mathcal{O}_L = \prod_{i=1}^{n} \mathcal{O}_L/B_i^{e_i} \rightsquigarrow \kappa(p)$-vector space, choose $a_j \in \mathcal{O}_L$, s.t. $\{a_j^{L}\}$ is a basis of $\mathcal{O}_L/B_i^{e_i}$ over $\kappa(p)$
$\Rightarrow$ $\{a_j\}$: basis of L/K
$\llcorner$ $[L:K] = \sum e_i f_i$

Shun/翔海(@shun4midx)

"$\Leftarrow$": If $\delta(\mathcal{O}_L/\mathcal{O}_K) \subseteq p$, then $\det(\mathrm{Tr}_{L/K}(a_j a_n^m)) = 0 \pmod p$

$\Rightarrow \prod \det(\mathrm{Tr}_{L/K}(a_j a_n^i)_{j,n}) = 0 \pmod p$

$\Rightarrow \exists i$, s.t. $\det(\mathrm{Tr}_{L/K}(a_j a_m^i)) = 0$

$\therefore \begin{cases} ① \ e_i > 1 \Rightarrow p: \text{ramified} \\ ② \ e_i = 1 \Rightarrow \mathcal{K}(B_i)/\mathcal{K}(p) \text{ insep} \Rightarrow p: \text{ramified} \end{cases}$ $\square$

"$\Leftarrow$": Given p: ramified, there are two cases:

① $e_i = 1 \ \forall i$, then $\mathcal{K}(B)/\mathcal{K}(p)$ insep for some i.

$\therefore \det(\mathrm{Tr}_{L/K}(a_j a_m^i)) = 0 \pmod p \Rightarrow p \mid \delta(\mathcal{O}_L/\mathcal{O}_K)$

② $\exists e_i > 1$ for some i, then $\exists\ x_i \neq 0$ in $\mathcal{O}_L/B_i^{e_i}$ that's nilpotent

$\forall a \in \mathcal{O}_L/p\mathcal{O}_L$, xa: nilpotent.

$\therefore \mathrm{Tr}_{L/K}(xa) = 0$ (m_{xa}: nilpotent)

$\therefore \mathrm{Tr}_{L/K}$: degenerate on $\mathcal{O}_L/p\mathcal{O}_L \Rightarrow \det(\mathrm{Tr}_{L/K}) = 0 \pmod p \Rightarrow p \mid \delta(\mathcal{O}_L/\mathcal{O}_K)$ $\square$

COROLLARY

There are only finitely many prime ideals in $\mathcal{O}_K$ ramified in $\mathcal{O}_L$

DECOMPOSITION GROUPS AND INERTIA GROUPS

SETTING

As before, $\mathcal{O}_K$: DDK, $K = \mathrm{Frac}(K)$, L/K: finite + sep, $\mathcal{O}_L$: integral closure of $\mathcal{O}_K$ in L

Suppose L/K is Gal, $G = \mathrm{Gal}(L/K)$

$B \trianglelefteq \mathcal{O}_L$: max, $p = B \cap \mathcal{O}_K$, $D(B|p) := \{\sigma \in G \mid \sigma(B) = B\}$

$G \curvearrowright \mathcal{O}_L \Rightarrow D(B|p) \curvearrowright \mathcal{O}_L/B$

$\therefore \exists$ group homo: $D(B|p) \xrightarrow{\pi} \mathrm{Aut}_{\mathcal{K}(p)}(\mathcal{K}(B)/\mathcal{K}(p))$

Define $I(B|p) := \ker(\pi)$, i.e. $\{\sigma \in D(B|p) \mid \sigma(a) = a \pmod B \ \forall a \in \mathcal{O}_L\}$

We want: $1 \to I(B|p) \to D(B|p) \xrightarrow{\pi} \mathrm{Aut}(\mathcal{K}(B)/\mathcal{K}(p))$

LEMMA

$\mathcal{K}(B)/\mathcal{K}(p)$ is a normal extension

<u>Proof</u>

$\forall$ irred $f(x) \in \mathcal{K}(p)[x]$, s.t. $a \in \mathcal{O}_L$, $f(\bar{a}) = 0$ in $\mathcal{K}(B)$

Let $\varphi_{a,K} \in \mathcal{O}_K[x]$: min poly of a over K

Then, L/K: Gal $\Rightarrow \varphi_{a,K}$ splits completely in L

$\Rightarrow \varphi_{a,K}$ splits completely in $\mathcal{O}_L$ ($\because a \in \mathcal{O}_L$)

$\Rightarrow \overline{\varphi_{a,K}} \in \mathcal{K}(p)[x]$ splits completely in $\mathcal{K}(B)$

$\therefore \bar{f}(x)$ splits completely in $\mathcal{K}(B)$ $\square$

REMARK

K: number field $\Rightarrow \mathcal{K}(B)/\mathcal{K}(p)$: Gal

FROBENIUS' THEOREM

$D(B|p) \to \mathrm{Aut}(\mathcal{K}(B)/\mathcal{K}(p))$: surjective

<u>Proof</u>

Given $g \in \mathrm{Aut}(\mathcal{K}(B)/\mathcal{K}(p))$,

Define (☆) to be $\exists \sigma \in G$, s.t. $\overline{\sigma(a)} = g(\bar{a}) \pmod B$ $\forall a \in \mathcal{O}_L$

Shun/翔海(@shun4midx)

① If σ satisfies (☆), $\forall a \in B$, $\sigma(a) \equiv 0 \pmod B$

$\therefore \sigma(B) \subseteq B \Rightarrow \sigma(B) = B$

② Otherwise, let $\{a_i\}_{i=1}^n \subseteq \mathcal{O}_L$ be a set of generators of $\mathcal{O}_L$ over $\mathcal{O}_K$

Define $h(Y, \underline{X_i}) := \prod_{\sigma \in G} (Y - \sum_{i=1}^n \sigma(a_i) X_i) \in \mathcal{O}_L[Y, \underline{X_i}]$

$\forall \sigma \in G$, $\sigma h(Y, \underline{X_i}) = h(Y, \underline{X_i}) \Rightarrow h \in \mathcal{O}_L^G[Y, \underline{X_i}]$

$\mathcal{O}_L^G = \mathcal{O}_K$, $\mathcal{O}_L \cap K$

$\mathcal{K}(p)[Y, \underline{X_i}]$

$\therefore \bar{h} = h \pmod p$ and $\bar{h}(\sum_{i=1}^n \bar{a}_i X_i, \underline{X_i}) = 0$

$\prod_{\sigma \in G} (\sum_i \bar{a}_i X_i - \sum_{i=1}^n g(\overline{\sigma(a_i)}) X_i)$

$\Rightarrow \forall g \in \mathrm{Aut}(\mathcal{K}(B)/\mathcal{K}(p))$, $g\bar{h} = \bar{h} \Rightarrow g\bar{h}(\sum \bar{a}_i X_i, \underline{X_i}) = 0$

$g\bar{h} = \prod_{\sigma \in G} (Y - \sum g(\overline{\sigma(a_i)}) X_i)$

$\Rightarrow \exists \sigma$, s.t. $\sum_{i=1}^n (\bar{a}_i - g(\overline{\sigma(a_i)}) X_i = 0$

$\Rightarrow \exists \sigma$, s.t. $\overline{\sigma(a_i)} = g^{-1}(\bar{a}_i)$

$\because \{a_i\}$: set of generators

$\therefore \overline{\sigma(a)} = g^{-1}(\bar{a})$ $\forall a \in \mathcal{O}_L$ □

CYCLOTOMIC FIELDS

SETTING

Let $\zeta_n \in \mathbb{C}$: n^{th} root of unity, $L = \mathbb{Q}(\zeta_n)$

RECALL

$\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Gal with $\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, $[L:\mathbb{Q}] = \phi(n)$, ϕ: Euler function

LEMMA

Say $n = p^s$, p: prime in $\mathbb{Z}$, let $\eta = \zeta_n - 1$. Then, we have

(i) $N_{L/\mathbb{Q}}(\eta) = (p)$

(ii) $p = u\eta^{\phi(p^s)}$ for some $u \in \mathbb{Z}[\zeta_n]^\times$

(iii) $\delta(\mathbb{Z}[\zeta_n]/\mathbb{Z}) = (p)^{sr - p^{s-1}}$, $r = [L:\mathbb{Q}] = \varphi(p^s)$

<u>Proof</u>

(i) Let Φ_n := min poly of ζ_n over $\mathbb{Q}$

Then, $\Phi_n = \dfrac{x^{p^s} - 1}{x^{p^{s-1}} - 1} = (x^{p^{s-1}})^{p-1} + \dots + x^{p^{s-1}} + 1$

Let f: min poly of η, $f = \Phi_n(x+1)$

Then, $N_{L/\mathbb{Q}}(\eta) = (\Phi_n(1)) = (p)$ ✓

(ii) Let $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$

Notice, $N_{L/\mathbb{Q}}(\eta) = \prod_{i, p \nmid i} (\zeta_n^i - 1) = p$ (N: field norm, Nr: ideal norm, i.e. size)

Claim: $\forall i$ s.t. $(i,p) = 1$, $\zeta_n^i - 1 \sim \zeta_n - 1$

<u>Proof</u>: Since $(i,p) = 1$, thus $\exists h$, s.t. $ih \equiv 1 \pmod{p^s} \Rightarrow \zeta_n = (\zeta_n^i)^h$

$\therefore \zeta_n - 1 = (\zeta_n^i)^h - 1 = (\zeta_n^i - 1)(1 + \zeta_n^i + \dots + (\zeta_n^i)^{h-1}) \Rightarrow \frac{\zeta_n - 1}{\zeta_n^i - 1} \in \mathbb{Z}[\zeta_n]^\times$, which is a unit since $\frac{\zeta_n^i - 1}{\zeta_n - 1} \in \mathbb{Z}[\zeta_n]^\times$ ✓

i s.t. $(i,p) \equiv 1 \pmod{p^s}$

As all factors in the norm are associates, thus $p = u(\zeta_n - 1)^{\varphi(p^s)}$

(iii) $\Phi_n'(\zeta_n) = p^s / \zeta(\zeta^{p^{s-1}} - 1)$

$\Rightarrow \delta(\mathbb{Z}[\zeta_n]/\mathbb{Z}) = N_{L/K}(\Phi_n'(\zeta_n))$ ... calculation □

Shun/翔海(@shun4midx)

REMARK

If $p^s > 2$, $sr - p^{s-1} > 0 \Rightarrow p \mid \delta(\mathbb{Z}[\zeta_n]/\mathbb{Z}) \leq \delta(\mathcal{O}_L/\mathbb{Z})$

PROPOSITION

$\mathcal{O}_L = \mathbb{Z}[\zeta_n]$

<u>Proof</u>

$p^d = \delta(\mathbb{Z}[\zeta_n]/\mathbb{Z}) = \delta(\mathcal{O}_L/\mathbb{Z})\,[\mathcal{O}_L : \mathbb{Z}[\zeta_n]]^2 \Rightarrow p^d \mathcal{O}_L \subseteq \mathbb{Z}[\zeta_n]$

Claim: $p\mathcal{O}_L \cap \mathbb{Z}[\zeta_n] = p\mathbb{Z}[\zeta_n]$

<u>Proof Sketch</u>

$\gcd(m,n)=1$, then:

$$\begin{tikzcd}
\mathbb{Q}(\zeta_{mn}) = \mathbb{Q}(\zeta_m)\,\mathbb{Q}(\zeta_n) \arrow[d, dash] & \\
L \arrow[dr, dash] & L \arrow[dl, dash] \\
& K
\end{tikzcd}$$

decomposition group (splits primes)

$$\Rightarrow \begin{cases} L^D \subseteq M \subseteq L, & M \subseteq L^D \\ L^I \subseteq M \subseteq L, & M \subseteq L^I \end{cases}$$

inertia group (ramifies primes)